Practical No. 14 – Chi-Square Test for Goodness of Fit

Blue = given data. Black = formulas. $\chi^2 = \Sigma\,(O-E)^2/E$. Solved by the critical-value method and the p-value method (and CHITEST as a check where df = k−1).

Ex.1 Accidents per day – goodness of fit to Poisson distribution

H0: accidents per day follow a Poisson distribution vs H1: they do not. λ is estimated from the data (1 parameter).

Accidents (k)	Observed (days)	Expected	(O−E)²/E	Note
0	34	29.4304	0.7095	
1	25	29.4304	0.6669	
2	11	14.7152	0.9380	
3	7	4.9051	0.8947	
4	3	1.5191	1.4438	*k = 4 means 4 or more*
Total days N	80	*Σ observed*		
Estimated λ (mean accidents/day)	1.0000	*λ = Σ k·O / N*		
χ^2 without pooling classes	4.6530	*for comparison only*		

Expected frequencies for k = 3 and k = 4 are below 5, so these two classes are pooled (rule: each E should be ≥ 5):

Pooled class	Observed	Expected	(O−E)²/E	Note
0	34	29.4304	0.7095	
1	25	29.4304	0.6669	
2	11	14.7152	0.9380	
3 or more	10	6.4241	1.9905	
Significance level α	0.05			
Test statistic χ^2	**4.3049**	*$\chi^2 = \Sigma\,(O-E)^2/E$*		
Degrees of freedom	2	*df = classes − 1 − 1 (λ estimated)*		
Critical value $\chi^2(\alpha, df)$	5.9915	*CHIINV(α, df)*		
p-value	0.11620	*CHIDIST(χ^2, df)*		
Decision (critical value method)	**Fail to reject H0**			
Decision (p-value method)	**Fail to reject H0**			

Conclusion: χ^2 is below the critical value (same result without pooling), so the data do not contradict a Poisson distribution.

Ex.2 Male/female births in 830 families of four – Binomial fit (n = 4, p = 0.5)

H0: number of male births ~ Binomial(4, 0.5) vs H1: it does not. (p given, so no parameter estimated)

Children per family n	4			
P(male birth) p	0.5			
Male births (k)	**Observed (families)**	**Expected**	**(O−E)²/E**	**Female births**
0	32	51.8750	7.6148	4
1	178	207.5000	4.1940	3
2	290	311.2500	1.4508	2
3	236	207.5000	3.9145	1
4	94	51.8750	34.2075	0
Total families	830	*Σ observed*		
Significance level α	0.05			
Test statistic χ^2	**51.3815**	*$\chi^2 = \Sigma\,(O-E)^2/E$*		
Degrees of freedom	4	*df = classes − 1*		
Critical value $\chi^2(\alpha, df)$	9.4877	*CHIINV(α, df)*		
p-value	0.00000	*CHIDIST(χ^2, df)*		
Excel CHITEST p-value (check)	0.00000	*CHITEST(observed, expected)*		
Decision (critical value method)	**Reject H0**			
Decision (p-value method)	**Reject H0**			

Conclusion: Reject H0 – data are not consistent with Binomial law with p = q = 0.5.

Ex.3 Die thrown 276 times – test of fairness (uniform distribution)

H0: die is unbiased (each face has probability 1/6) vs H1: die is biased.

Face	Observed	Expected	(O−E)²/E	
1	40	46.0000	0.7826	
2	32	46.0000	4.2609	
3	29	46.0000	6.2826	
4	59	46.0000	3.6739	
5	57	46.0000	2.6304	
6	59	46.0000	3.6739	
Total throws	276	*Σ observed*		
Significance level α	0.05			
Test statistic χ^2	**21.3043**	*$\chi^2 = \Sigma\,(O-E)^2/E$*		
Degrees of freedom	5	*df = classes − 1*		
Critical value $\chi^2(\alpha, df)$	11.0705	*CHIINV(α, df)*		
p-value	0.00071	*CHIDIST(χ^2, df)*		
Excel CHITEST p-value (check)	0.00071	*CHITEST(observed, expected)*		
Decision (critical value method)	**Reject H0**			
Decision (p-value method)	**Reject H0**			

Conclusion: Reject H0 – evidence is not enough to support that the die is unbiased.

Ex.4 Popularity of four movies – test of equal popularity

H0: all four movies are equally popular (each 1/4) vs H1: popularity differs.

Movie	Customers (observed)	Expected	(O−E)²/E	
1	86	82.0000	0.1951	
2	77	82.0000	0.3049	
3	84	82.0000	0.0488	
4	81	82.0000	0.0122	
Total customers	328	*Σ observed*		
Significance level α	0.05			
Test statistic χ^2	**0.5610**	*$\chi^2 = \Sigma\,(O-E)^2/E$*		
Degrees of freedom	3	*df = classes − 1*		
Critical value $\chi^2(\alpha, df)$	7.8147	*CHIINV(α, df)*		
p-value	0.90531	*CHIDIST(χ^2, df)*		
Excel CHITEST p-value (check)	0.90531	*CHITEST(observed, expected)*		
Decision (critical value method)	**Fail to reject H0**			
Decision (p-value method)	**Fail to reject H0**			

Conclusion: Fail to reject H0 – no evidence of a difference in popularity of the four movies.